

Vulnerability Assessment Report

for a Live Website

Target: scanme.nmap.org

Date: 20 May 2026

Prepared by: Maramakshaya

Report Details

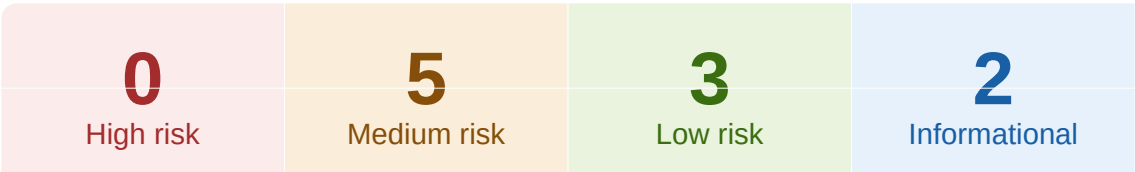
Target Details

Author	Maramakshaya	Target URL	scanme.nmap.org
Date	20 May 2026	IP Address	45.33.32.156
Report Type	Vulnerability Assessment	OS Detected	Linux (Ubuntu)
Classification	Confidential	Scan Tool	Nmap 7.98 + OWASP ZAP 2.17.0

Executive summary

A comprehensive vulnerability assessment was conducted on **scanme.nmap.org** using Nmap 7.98 and OWASP ZAP 2.17.0. The assessment identified open network ports, running services, and 10 web application security vulnerabilities classified by risk level. All work is documented in the public GitHub repository: FUTURE_CS_01.

Risk summary



Tools used

Nmap 7.98	Network port scanner — service and OS version detection
OWASP ZAP 2.17.0	Web application security scanner — passive and active scanning

Nmap scan findings

Nmap scan screenshot

```
(root@kali)-[/home/akiii]
# nmap -sV -sC scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-18 20:29 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.29s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 992 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   1024 ac:00:a0:1a:82:ff:cc:55:99:dc:67:2b:34:97:6b:75 (DSA)
|   2048 20:3d:2d:44:62:2a:b0:5a:9d:b5:b3:05:14:c2:a6:b2 (RSA)
|   256  96:02:bb:5e:57:54:1c:4e:45:2f:56:4c:4a:24:b2:57 (ECDSA)
|_  256  33:fa:91:0f:e0:e1:7b:1f:6d:05:a2:b0:f1:54:41:56 (ED25519)
25/tcp    filtered smtp
80/tcp    open  http         Apache httpd 2.4.7 ((Ubuntu))
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-title: Go ahead and ScanMe!
|_ http-favicon: Nmap Project
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
9929/tcp  open  nping-echo   Nping echo
31337/tcp open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 47.03 seconds
```

Open ports and services

Port	State	Service	Version	Risk
22/tcp	Open	SSH	OpenSSH 6.6.1p1 Ubuntu	Medium
25/tcp	Filtered	SMTP	—	Info
80/tcp	Open	HTTP	Apache httpd 2.4.7 (Ubuntu)	Medium
9929/tcp	Open	Nping echo	Nping echo	Info
31337/tcp	Open	tcpwrapped	—	Low

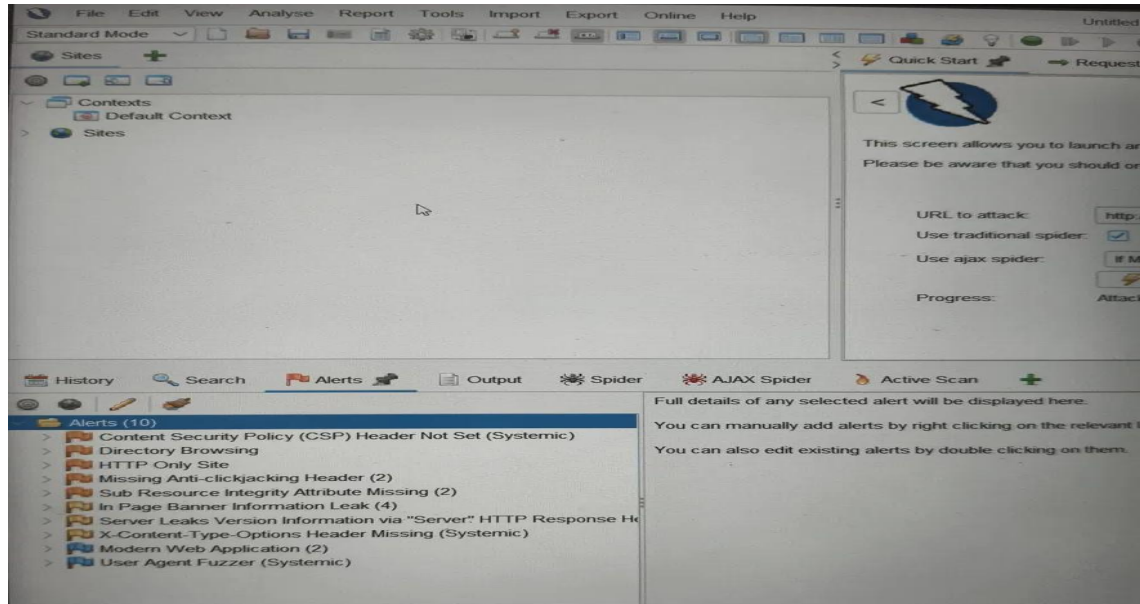
Key observations

- OpenSSH 6.6.1p1 is outdated (released 2014) and contains known CVEs.
- Apache 2.4.7 is outdated — multiple security patches have been released since.
- Port 31337 is historically associated with malicious use and should be monitored.
- No HTTPS detected on port 443 — all web traffic is unencrypted.

SECTION 3

OWASP ZAP findings

ZAP alerts screenshot



Medium risk vulnerabilities

- 1 Content Security Policy (CSP) header not set** **Medium**

No CSP header found — allows attackers to inject malicious scripts (XSS attacks).

Fix: Add Content-Security-Policy header in Apache/Nginx server configuration.
- 2 Directory browsing enabled** **Medium**

Server exposes directory listings — attackers can view sensitive files and directories.

Fix: Disable directory listing with Options -Indexes in Apache configuration.
- 3 HTTP only site (no HTTPS)** **Medium**

All traffic sent in plain text — susceptible to man-in-the-middle (MITM) attacks.

Fix: Install SSL/TLS certificate and configure redirect from HTTP to HTTPS.

4	Missing anti-clickjacking header	Medium
No X-Frame-Options header present — site can be embedded in iframes for clickjacking attacks.		
<i>Fix: Add X-Frame-Options: DENY or SAMEORIGIN to server response headers.</i>		
5	Sub resource integrity attribute missing	Medium
External scripts loaded without integrity checks — risk of supply chain attacks.		
<i>Fix: Add integrity and crossorigin attributes to all external script and link tags.</i>		

Low risk and informational findings

Low risk vulnerabilities

- | | | |
|---|--|------------|
| 6 | In-page banner information leak | Low |
| <p>Server banner reveals software version details useful to attackers for targeted exploits.</p> <p><i>Fix: Configure <code>ServerTokens Prod</code> in Apache to suppress detailed version info.</i></p> | | |
| <hr/> | | |
| 7 | Server leaks version via HTTP response header | Low |
| <p>Apache version disclosed in HTTP response headers — aids targeted attack planning.</p> <p><i>Fix: Set <code>ServerSignature Off</code> and <code>ServerTokens Prod</code> in Apache configuration.</i></p> | | |
| <hr/> | | |
| 8 | X-Content-Type-Options header missing | Low |
| <p>Browser may MIME-sniff responses leading to content injection or XSS attacks.</p> <p><i>Fix: Add <code>X-Content-Type-Options: nosniff</code> to all server responses.</i></p> | | |

Informational findings

- | | | |
|---|--|-------------|
| 9 | Modern web application detected | Info |
| <p>Site uses modern JavaScript frameworks — noted for reference and further review.</p> | | |
| <hr/> | | |
| 10 | User agent fuzzer | Info |
| <p>Server responds differently to various user agent strings — may indicate inconsistent configuration.</p> | | |

Conclusion and recommendations

The vulnerability assessment of **scanme.nmap.org** revealed no critical or high risk vulnerabilities. However, 5 medium risk issues related to missing security headers and unencrypted HTTP traffic require immediate attention to improve the security posture of the application.

Priority remediation steps

Immediate — Medium risk

- Enable HTTPS with SSL/TLS certificate
- Add Content-Security-Policy header
- Add X-Frame-Options: DENY header
- Disable directory browsing
- Add SRI to all external scripts

Short term — Low risk

- Hide Apache server version info
- Add X-Content-Type-Options: nosniff
- Update OpenSSH to latest version
- Update Apache to latest version
- Monitor and review port 31337

GitHub Repository	https://github.com/maramakshaya-code/FUTURE_CS_01
-------------------	---

ZAP HTML Report	https://1drv.ms/u/c/CD48CF292C9DAF6F/IQCO6D5nt0lrSreBEu2Vjw9zAUgm6AnHB1j-oYCGIFTdBR4
-----------------	---

Prepared by	Maramakshaya
-------------	--------------

Date	20 May 2026
------	-------------

Tools	Nmap 7.98 OWASP ZAP 2.17.0
-------	------------------------------